Empirical and Molecular Formulas

Simplest ratios, formulas from percentages, and molecular formulas

Name _____ Class _____ Date _____

DATA

H = 1.008 g/mol C = 12.01 g/mol N = 14.01 g/mol O = 16.00 g/mol Ca = 40.08 g/mol Fe = 55.85 g/mol

Show your working: % → g → mol → divide by the smallest → whole numbers.

A. Understanding

1. What is the difference between an empirical formula and a molecular formula?

2. Write the empirical formula of: (a) C_6H_6, (b) C_2H_4, (c) H_2O_2, (d) C_8H_{18}, (e) N_2O_4.

3. After dividing by the smallest number of moles, a student gets a ratio of 1 : 1.50 and writes the formula as AB_2. What is wrong, and what is the correct ratio?

B. Empirical formulas from percent composition

4. A hydrocarbon is 85.63% carbon and 14.37% hydrogen by mass. Find its empirical formula.

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

5. A compound is 40.04% calcium, 12.00% carbon and 47.96% oxygen by mass. Find its empirical formula.

6. An oxide of iron is 72.36% iron and 27.64% oxygen by mass. Find its empirical formula.

C. Molecular formulas

7. A compound has empirical formula CH_2O and molar mass 60.05 g/mol. Find its molecular formula.

8. Acetylene has empirical formula CH and molar mass 26.04 g/mol. Find its molecular formula.

D. Challenge

9. A compound of nitrogen and oxygen is 30.43% nitrogen by mass, and its molar mass is 92.02 g/mol. Find its empirical and molecular formulas.

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.